Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

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Motivation and problem

Motivation

- Understand why teachers leave the profession.
- Is there a real policy margin to soften teacher shortages by retaining teachers — or bringing them back?

Problem

- Official microdata on teachers are not publicly available for many countries.
- We start with the United States: the country with the most information available.

Solution

Use the Current Population Survey to

A. reproduce and extend the official statistics on teachers leaving the profession;

B. having achieved A, study in detail *how many* leave, *where* they go, and *why*.

0

Methodological discussion

The Current Population Survey

- The monthly household survey of the United States, run by the Census Bureau for the Bureau of Labor Statistics. Nationally representative of the civilian non-institutional population.
- We pool 28 consecutive March supplements, 1998–2025: **5.0 million** person-year records.
- **104,545** of them are teachers — about 3,700 per year.

unemployment interviews migration identity

The March supplement (ASEC)

- Run by the Census Bureau each March since 1948: the source of the official annual income, poverty and health-insurance statistics.
- About 90,000 households: the March sample plus oversamples interviewed in February and April.
- Every adult reports on the *previous calendar year*: longest job held, weeks worked, earnings, benefits.

Identifying a teacher: the census occupation codes

	1990 classification surveys 1998–2002	2002/2010 classification surveys 2003–2019	2018 classification surveys 2020–2025
Pre-K and kindergarten	155	2300	2300
Elementary (and middle)	156	2310	2310
Secondary	157	2320	2320
Special education	158	2330	2330
Other teachers and instructors	159	2340	2360

The K–12 block maps one-to-one across the three regimes: the *group* is constant over time even where single codes are not.

▸ traceability


interviewed 4 months
8 months out
interviewed 4 months
1
2
3
4
5
6
7
8
Mar
Jun
Mar
Jun
year 1
year 2
the same interview, twelve months later

Two ways to identify a leaver

A. Follow the individual

All months. Links require matching:

household identifier
dwelling identifier
person line number

validated on sex, race, age +0–2

B. Ask about last year

The March supplement, retrospectively.

Taught as last year's longest job, not teaching this March $\Rightarrow$ left.

Disadvantages of each option

A. Following individuals

1. Identity is not certain: validation can admit noisy people.
2. Movers are lost: $\sim$25% of addresses are not found a year later.
3. Normal household attrition on top.
4. Strong seasonality in teaching.
5. Many ways to define the rate (next slide).

▸ linkage loss ▸ attachment ▸ validation

B. Retrospective March

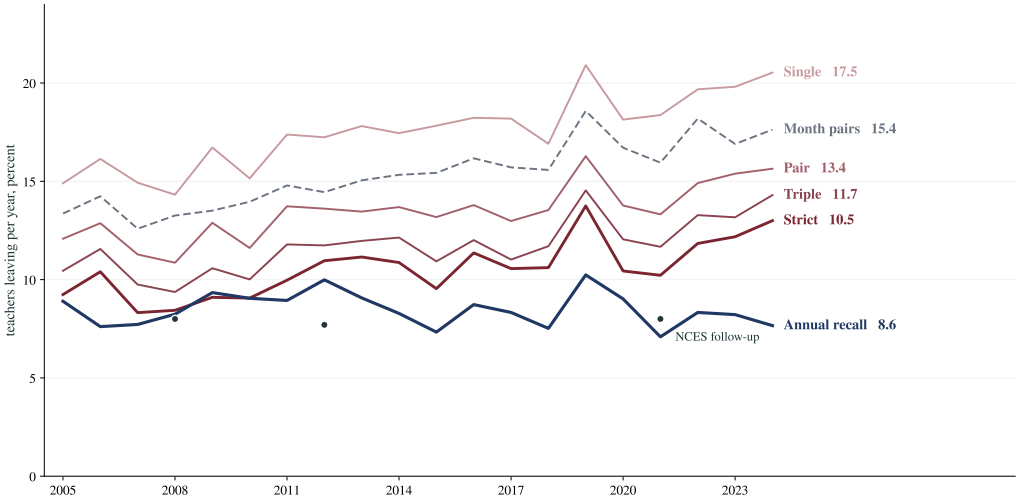
1. Recall bias.
2. Less sample: one observation per person per year ($\sim$3,700 teachers) against $\sim$11,000 monthly teacher observations.
3. Only March-to-March transitions; within-year spells are smoothed away.

Six ways to define leaving, simplest to strictest

Unit: the individual. Four interviews in year 1, four in year 2.

	A teacher:	A leaver:
Single	seen teaching in $\geq$ 1 of the 4 year-1 interviews	seen teaching in none of the 4 year-2 interviews
Pair	seen teaching in $\geq$ 2 of the 4	seen teaching in none of the 4
Triple	seen teaching in $\geq$ 3 of the 4	seen teaching in none of the 4
Strict	seen teaching in all 4	seen teaching in none of the 4
Month pairs	each teaching month, on its own	not seen teaching twelve months later
Annual recall	teaching was last year's longest job	not teaching at the March interview

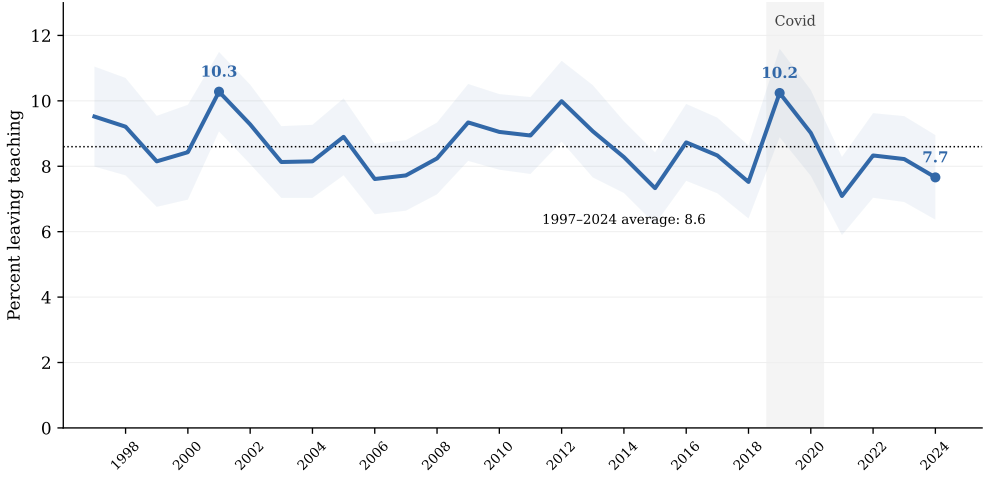
The six, on one axis



1

Leaving rate: how many leave

Leaving teaching, 1997–2024



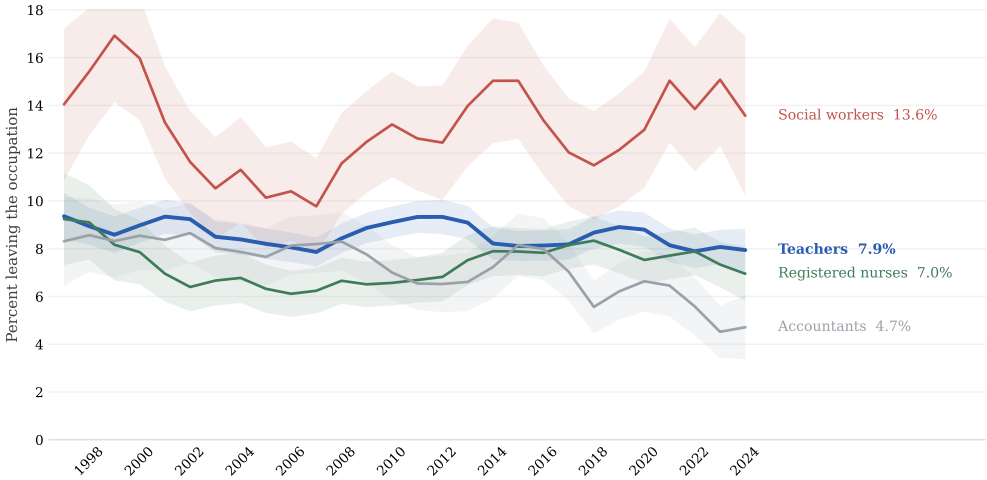
▸ public vs private

Comparable professions: who we compare teachers with

	Teachers	Nurses	Social workers	Accountants
Female (%)	76.0	88.2	82.9	57.0
Average age (years)	44.4	43.1	43.4	44.1
Master's or higher (%)	51.9	25.9	50.8	31.3
Public sector (%)	70.2	12.3	43.1	11.0
Weeks worked	46.6	49.3	49.6	49.5
Median annual wage ($)	56,000	80,000	57,000	77,000
Observations	28,932	13,107	3,407	6,897

College graduates, pooled 2015–2024, weighted; wages from 2020–2024.

Leaving the occupation, by profession

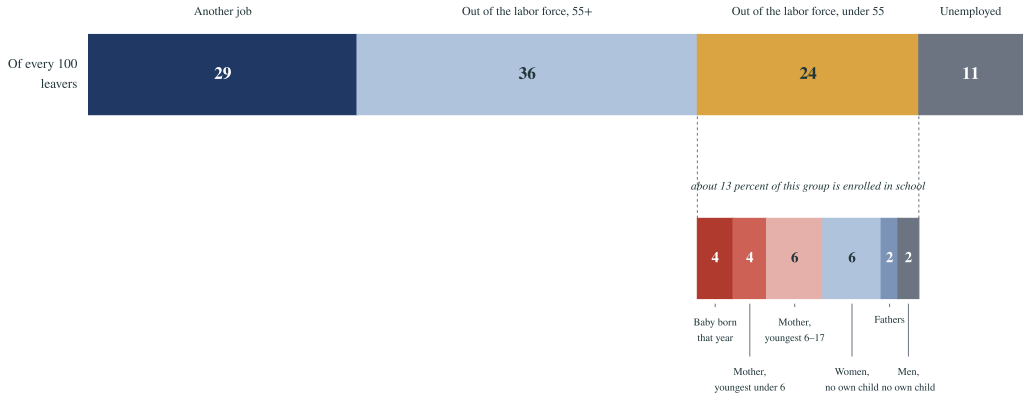


▸ more professions

2

Composition: where they go

Of every 100 leavers (average, 2020–2024)



Message. A third find another job; a third retire; the rest split between early labor-force exit and unemployment.

▸ vs NCES follow-up

Where the employed leavers go

EDUCATION, NON-TEACHING — 23%

- 8.1 Postsecondary teachers
- 4.3 Tutors
- 4.2 Teaching assistants
- 3.1 Education administrators
- 1.5 Librarians
- 1.3 Other instruction and library wo...
- 0.2 Archivists, curators, and museum...

CARE — 9%

- 1.8 Educational and career counselors
- 1.2 Social and human service assista...
- 1.1 Mental health and substance abus...
- 1.1 Social workers, all other
- 0.8 Childcare workers
- 0.6 Mental health counselors
- 0.5 Counselors, all other
- 0.5 Substance abuse and behavioral d...
- 0.4 Other community and social servi...
- 0.4 Probation officers and correctio...

PROFESSIONAL & MANAGERIAL — 32%

- 9.7 Managers, all other
- 1.9 Coaches and scouts
- 1.7 Architects
- 1.5 General and operations managers
- 1.3 Business operations specialists...
- 1.2 Other mathematical science occup...
- 1.1 Writers and authors
- 0.9 Broadcast, sound, and lighting t...
- 0.8 Chemical technicians
- 0.7 Social and community service man...
- 0.6 Property, real estate, and commu...
- 0.6 Photographers
- 0.6 Actors
- 0.6 Therapists, all other
- 0.5 Lodging managers
- 0.5 Construction managers
- 0.5 Miscellaneous health technologis...
- 0.4 Other psychologists
- 0.4 Other engineering technologists...
- 0.4 Management analysts
- 0.4 Market research analysts and mar...
- 0.4 Human resources workers
- 0.4 Veterinarians
- 0.4 Chief executives
- 0.4 Dietitians and nutritionists
- 0.4 Information security analysts
- 0.3 Claims adjusters, appraisers, ex...
- 0.3 Physical therapists
- 0.3 Food service managers
- 0.3 School psychologists
- 0.3 Editors
- 0.3 Transportation, storage, and dis...
- 0.2 Meeting, convention, and event p...
- 0.2 Training and development managers
- 0.2 Accountants and auditors
- 0.2 Other healthcare practitioners a...
- 0.2 Graphic designers
- 0.2 Training and development special...
- 0.2 Materials engineers
- 0.2 Civil engineers
- 0.1 Other designers
- 0.1 Wholesale and retail buyers, exc...
- 0.1 Physical scientists, all other
- 0.1 Project management specialists
- 0.1 Dietetic technicians and ophthal...
- 0.0 Floral designers
- 0.0 Farmers, ranchers, and other agr...
- 0.0 Landscape architects
- 0.0 Miscellaneous social scientists...

SALES, OFFICE & MANUAL — 36%

- 2.8 Secretaries and admin. assistants
- 1.8 Retail salespersons
- 1.8 Landscaping and groundskeeping w...
- 1.6 Exercise trainers
- 1.4 Supervisors of personal care and...
- 1.3 Taxi drivers
- 1.3 Police officers
- 1.2 Executive secretaries
- 1.1 Stockers and order fillers
- 1.0 Other life, physical, and social...
- 1.0 First-Line supervisors of non-re...
- 0.9 Title examiners, abstractors, an...
- 0.8 Personal care and service worker...
- 0.7 Fast food and counter workers
- 0.7 First-Line supervisors of retail...
- 0.7 Couriers and messengers
- 0.7 Statistical assistants
- 0.7 Paralegals and legal assistants
- 0.6 Cashiers
- 0.6 Recreation workers
- 0.6 First-line supervisors of landsc...
- 0.6 Legal support workers, all other
- 0.6 Other healthcare support workers
- 0.6 Laborers and freight, stock, and...
- 0.6 Animal caretakers
- 0.5 Insurance claims and policy proc...
- 0.5 Personal care aides
- 0.5 Bill and account collectors
- 0.4 Waiters and waitresses
- 0.4 Data entry keyers
- 0.4 Hairdressers, hairstylists, and...
- 0.4 Real estate brokers and sales ag...
- 0.4 Customer service representatives
- 0.4 First-Line supervisors of office...
- 0.4 Manicurists and pedicurists
- 0.4 Other entertainment attendants a...
- 0.4 Construction laborers
- 0.4 Office and administrative suppor...
- 0.3 Cooks
- 0.3 Bookkeeping, accounting, and aud...
- 0.3 Miscellaneous agricultural worke...
- 0.3 Travel agents
- 0.3 Sewing machine operators
- 0.3 Bartenders
- 0.3 Conveyor, dredge, and hoist and...
- 0.3 Boilermakers
- 0.3 Billing and posting clerks
- 0.3 Occupational health and safety s...
- 0.3 Packaging and filling machine op...
- 0.2 Laundry and dry-cleaning workers
- 0.2 Physical therapist assistants an...
- 0.2 Janitors and building cleaners
- 0.2 Payroll and timekeeping clerks
- 0.2 Nursing assistants
- 0.2 Helpers--production workers
- 0.1 Hosts and hostesses, restaurant...
- 0.1 Sales representatives, wholesale...
- 0.1 Security guards and gambling sur...
- 0.1 Chefs and head cooks
- 0.1 Receptionists and information cl...
- 0.1 Computer, automated teller, and...
- 0.1 Maintenance and repair workers...
- 0.1 Office clerks, general
- 0.0 Stationary engineers and boiler...
- 0.0 Tellers
- 0.0 Ushers, lobby attendants, and ti...

What this says for policy

- Of every 100 leavers, about 36 retire: a margin for *delaying* exit, not for return.
- 29 move to another job — half of them to destinations that do not require a degree.
- 11 are unemployed and 24 leave the labor force before 55; about 13% of the latter are studying.
- The retrievable margin is real but bounded: policy competes with retirement, family time and re-training — not only with other employers.

3

Correlations: why they leave

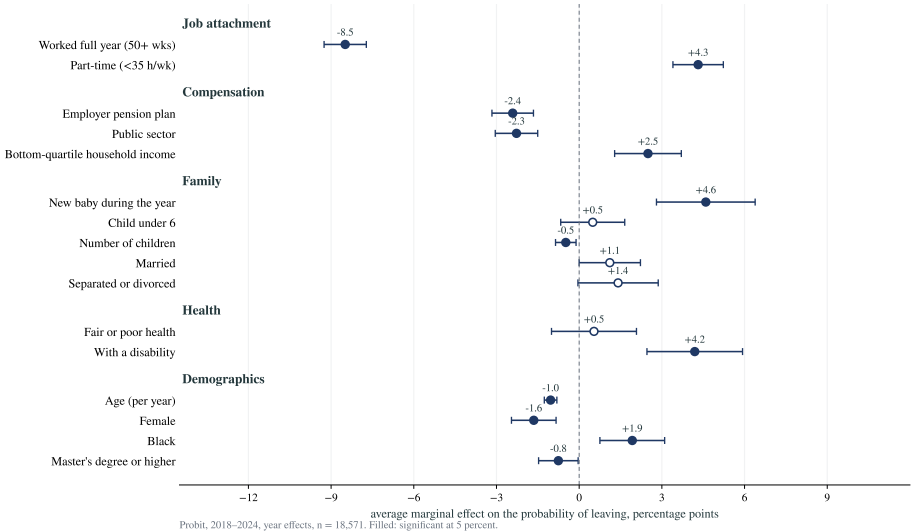
Specification

$$y_{it} = \alpha + \mathbf{a}'_{it}\beta_1 + \mathbf{c}'_{it}\beta_2 + \mathbf{f}'_{it}\beta_3 + \mathbf{h}'_{it}\beta_4 + \mathbf{d}'_{it}\beta_5 + \delta_t + \varepsilon_{it}$$

Probit; average marginal effects reported. 2018–2024, $n = 18{,}571$.

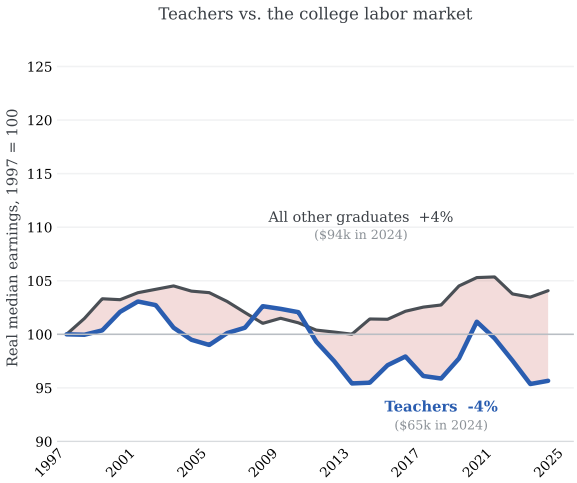
▸ LPM ▸ logit / probit ▸ bounded? ▸ what we observe

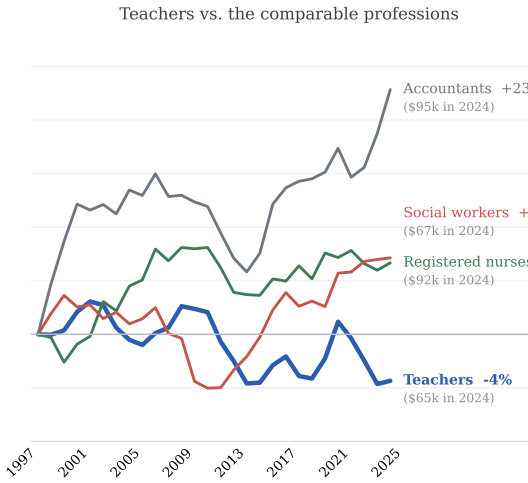
What correlates with leaving



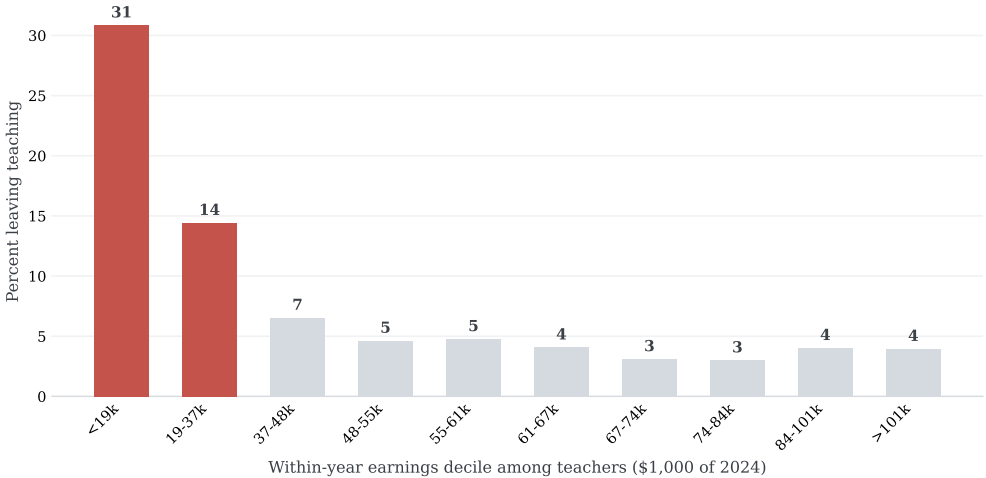
Probit, 2018–2024, year effects, n = 18,571. Filled: significant at 5 percent.

Relative pay since 1997

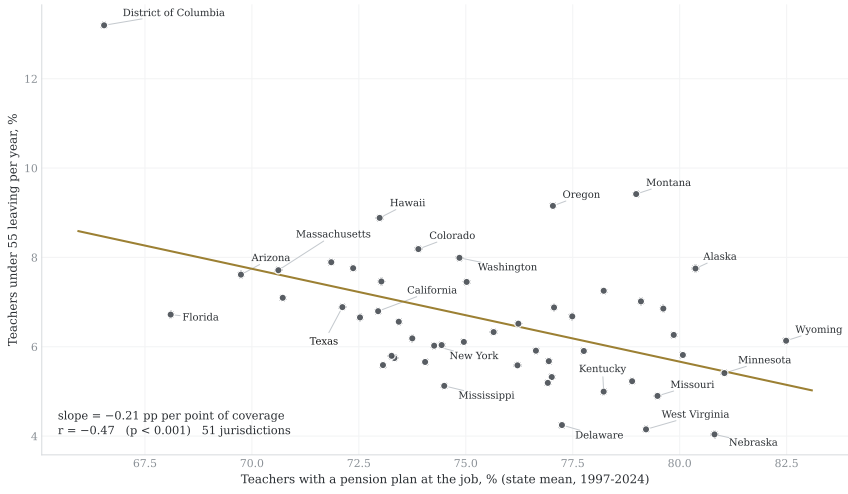




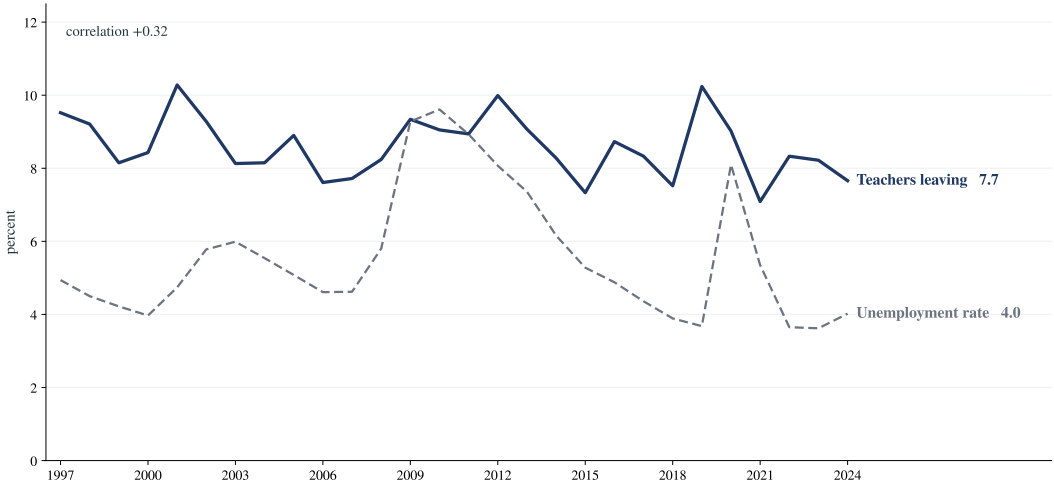
Leaving, by position in the teacher pay distribution



Pension coverage and leaving, across states



Leaving and the cycle



Conclusions

- A household survey can reproduce the official picture of teacher attrition — and extend it yearly, with covariates records do not have.
- Who leaves: the weakly attached, the low-paid, those without pension coverage, new mothers, teachers with a disability.
- We observe only part of the motives — financial and life-cycle, not working conditions — so these are margins, not the full story.
- What could bring teachers back: flexibility, economic security, pension coverage, and re-entry paths for those who left to study or care.

A

Appendix

How the CPS measures unemployment

- The *basic monthly* interview measures labor force status in the reference week — that alone produces the official monthly unemployment rate. Not the March supplement.
- The rest of the survey: monthly labor statistics (employment, hours, earnings) and topical supplements (school enrollment in October, voting in November, tobacco, computer use).
- The March supplement adds the annual income block: the source of income, poverty and health-insurance statistics — and of our retrospective measure.

back

Are interviews monthly? Who is new each month?

Yes: every month, one eighth of the sample sits in each position of the 4-8-4 rotation. Verified in the microdata:

Month in sample	1	2	3	4	5	6	7	8
March 2023 (%)	12.5	12.9	12.8	12.5	12.2	12.4	12.3	12.3
July 2023 (%)	12.3	12.9	12.9	12.5	12.2	12.2	12.5	12.4
January 2024 (%)	12.4	12.9	12.9	13.1	11.9	12.0	12.1	12.6

So each month: $\sim$12.5% new households, $\sim$37.5% in months 2–4, $\sim$50% returning in months 5–8. Month-to-month overlap 75%; year-to-year overlap 50%, by design.

back

Internal migration in the United States

- In 2023, **7.8%** of Americans changed residence within the year — 25.6 million people, the lowest rate on record.
- 3.2% moved to a different county; 1.4% to a different state.
- This is the loss channel of any address-based panel: linkage follows dwellings, not people.

Census Bureau, CPS ASEC, historical table A-1 (1948–2023).

back

Is the same person validated year to year?

- In the field, yes: the interviewer carries the household roster forward, each member keeps a line number, and occupation uses dependent interviewing between consecutive months.
- The *public* files carry no person identifier. We reconstruct the roster: household id + dwelling id + line number, validated on sex, race, and age advancing 0–2 years.
- Validation is not watertight: a mover replaced by someone of the same sex and similar age can slip through.

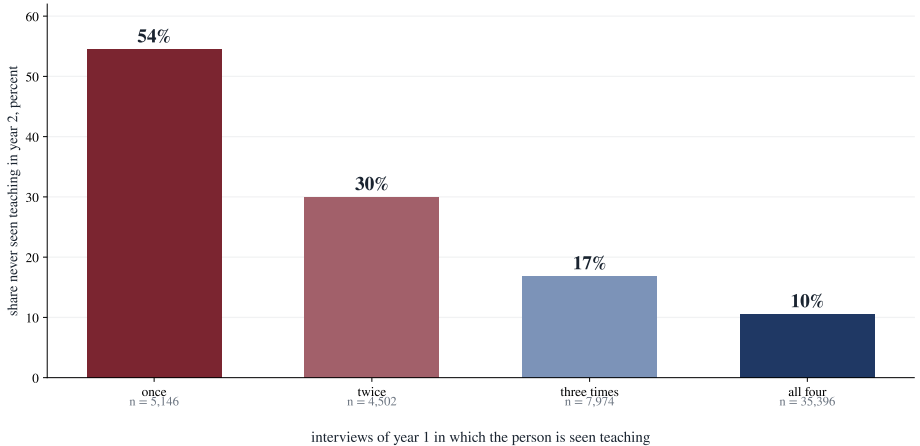
back

How much is lost in linking, and who

- Of the addresses due to return twelve months later, **74.5%** are found again (March 2023 $\rightarrow$ March 2024); the missing quarter is moves plus household non-response.
- Teachers link *more* than others, not less: $\sim$80% of teacher observations vs $\sim$76% of other college graduates.
- The loss is still selective: movers leave teaching more, so panel measures are lower bounds on mobility-related exit.

back

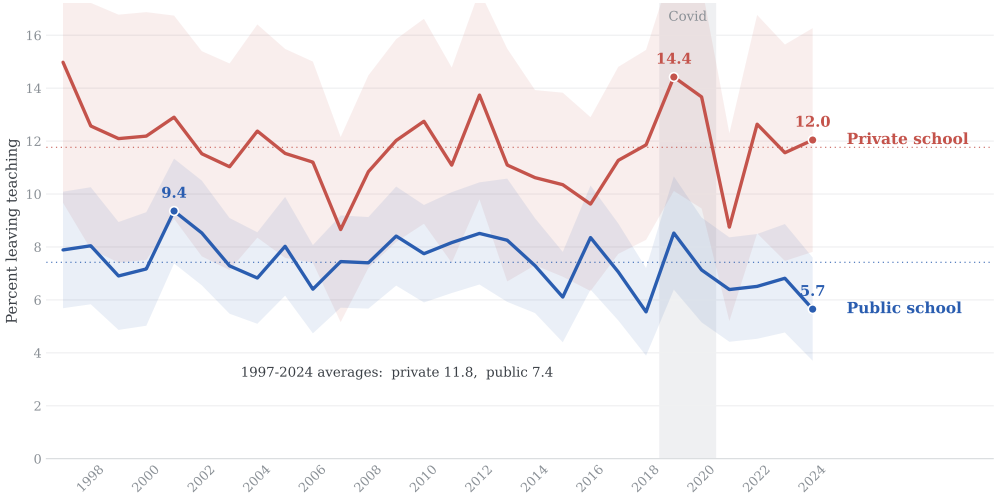
Attachment: seen once, gone



Of teachers seen teaching in a single year-1 interview, 54% are never seen teaching in their year-2 interviews; 10% of those seen in all four. Summer-only patterns are just 3% of these leavers.

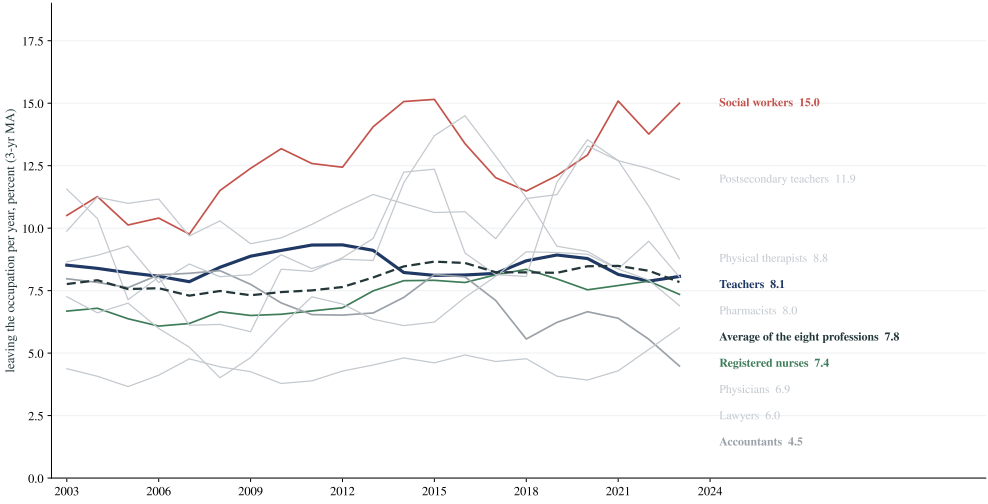
back

Public vs private schools



back

More professions, and their average



back

Our destinations vs the NCES follow-up survey

Of every 100 leavers…	CPS, 2020–2024	TFS, 2021–22
Working in another job	28.8	57.2
of which outside education	—	11.8
Retired / out of the LF, 55+	36.4	27.7
Out of the LF, under 55	23.7	10.1[a]
Unemployed	11.1	4.8

[a]Caring for family 5.6 + student 1.8 + disabled 0.3 + other 2.4. TFS covers regular classroom teachers in public schools and counts K–12 non-teaching jobs (39.3) as work; our occupation-based measure spans public and private and all teaching attachment. Sources: NCES 2024-039, table A-8 and NTPS estimate tables.

back

Is the teacher group constant across code changes?

- The five-code K–12 block maps one-to-one across the 1990, 2002/2010 and 2018 classifications; we always use the block, never a single code across regimes.
- Evidence: re-computing the whole series on the narrow core (dropping special education and other teachers) moves the pooled rate from 8.67 to 8.57 — a 0.1 point change.
- The 2003 and 2020 reclassifications leave no visible break in the series.

back

Linear probability model, fixed-effects variants

	Year FE	No FE	State + year FE
Worked full year	−11.6 (0.7)	−11.5 (0.7)	−11.7 (0.7)
Part-time	+8.7 (1.1)	+8.7 (1.1)	+8.6 (1.1)
Employer pension plan	−2.3 (0.5)	−2.4 (0.5)	−2.3 (0.5)
Public sector	−2.4 (0.6)	−2.4 (0.6)	−2.4 (0.6)
New baby	+5.6 (1.6)	+5.7 (1.6)	+5.7 (1.6)
With a disability	+8.9 (2.1)	+8.9 (2.1)	+8.9 (2.1)
Bottom-quartile income	+4.1 (1.2)	+4.2 (1.2)	+4.0 (1.2)

Coefficients $\times$ 100, robust SE in parentheses. 2018–2024, $n = 18{,}571$. Full set of controls as in the main specification.

back

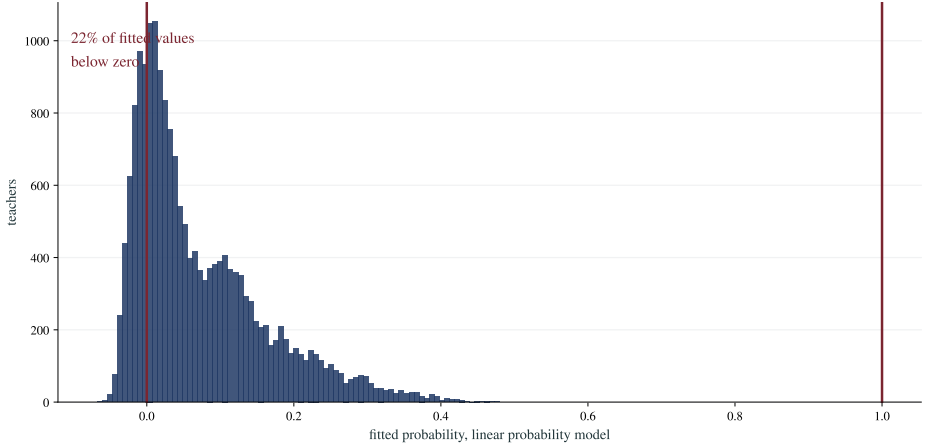
LPM vs logit vs probit

	LPM	Logit AME	Probit AME
Worked full year	−11.6	−11.0	−10.8
Part-time	+8.7	+4.4	+5.0
Employer pension plan	−2.3	−2.3	−2.4
Public sector	−2.4	−2.6	−2.6
New baby	+5.6	+6.1	+5.6
With a disability	+8.9	+5.3	+5.9
Bottom-quartile income	+4.1	+2.4	+2.6

Percentage points. Same specification and sample throughout. The ranking and significance of every margin is unchanged across estimators.

back

Is the LPM bounded between 0 and 1?



No: 22% of fitted values fall below zero. By that check, the probit is the main specification and the LPM a robustness variant.

back

What the CPS observes, of why teachers leave

Reasons in the literature	In the CPS?
Working conditions	no
Organizational support	no
Autonomy and intrinsic motives	no
Financial factors	**yes**
Personal life-cycle motives	**yes**
Sociodemographics	**yes**

back